

Thesis Progress Report

Thesis Topic: Benchmarking Holistic vs Sentence-Level RLAIIF for Factual Summarization via Granular Credit Assignment (GCA)

Date: 28 April 2026

1. Core Explanation of the Architecture

The method does not prefer a summary because it has fewer or more sentences. The comparison is based on factual claims supported by the article. The judge checks which summary is more factually consistent with the article, and GCA aggregates the local decisions into one summary-level preference.

```
Article + Summary A + Summary B
↓
Break summaries into sentences or claims
↓
Check each sentence or claim against the article
↓
LLM judge gives local factual judgments
↓
GCA aggregates local judgments
↓
Final preference: A, B, or no preference
↓
Use only clear preferences for DPO training
```

GCA comes after the local sentence-level or claim-level judgments. It is the aggregation layer that converts local evidence into one chosen/rejected pair that can be used for DPO training.

2. Example 1: Summary A has 4 sentences and Summary B has 6 sentences

2.1 Article

A German electric vehicle company, VoltAuto, announced on Monday that it will build a new battery factory in Mainz.

The factory will cost €2 billion and is expected to create 3,000 jobs.

Construction will begin in 2026, but the project still needs approval from environmental regulators.

The company said production could start in 2028 if approval is granted.

The important factual claims in the article are:

- F1: VoltAuto announced a new battery factory.
- F2: The factory will be in Mainz.
- F3: The announcement happened on Monday.
- F4: The factory will cost €2 billion.
- F5: The project is expected to create 3,000 jobs.
- F6: Construction will begin in 2026.
- F7: The project still needs environmental approval.
- F8: Production could start in 2028 if approval is granted.

2.2 Summary A

A1: VoltAuto announced on Monday that it will build a new battery factory in Mainz.
A2: The project will cost €2 billion and is expected to create 3,000 jobs.
A3: Construction is planned to begin in 2026.
A4: Production could start in 2028 if environmental approval is granted.

Summary A covers F1, F2, F3, F4, F5, F6, F7, and F8 without an obvious factual contradiction.

2.3 Summary B

B1: VoltAuto announced a new battery factory in Mainz.
B2: The company said the factory will cost €2 billion.
B3: The project is expected to create 3,000 jobs.
B4: Construction will begin in 2026.
B5: Environmental regulators have already approved the project.
B6: Production is guaranteed to start in 2028.

Summary B contains many correct facts, but it has two factual problems:

- B5 is wrong because the article says the project still needs environmental approval, while B says regulators have already approved it.
- B6 is too strong because the article says production could start in 2028 if approval is granted, while B says production is guaranteed to start in 2028.

3. Workflow for Example 1

3.1 Candidate summaries are generated

The system starts with one article and two candidate summaries: Summary A and Summary B. Both the holistic pipeline and the GCA pipeline use the same candidate pool, so the comparison remains fair.

3.2 Sentence segmentation

Summary A: A1, A2, A3, A4
Summary B: B1, B2, B3, B4, B5, B6

Sentence count is not the decision factor. Sentence splitting only makes the judgment more local and easier to inspect.

3.3 Semantic sentence alignment

Because Summary A has 4 sentences and Summary B has 6 sentences, direct index-based alignment would be weak. The alignment should be based on meaning.

A1 ↔ B1
A2 ↔ B2 + B3
A3 ↔ B4
A4 ↔ B5 + B6

A2 contains the cost and jobs information, while B splits these facts into two sentences. A4 contains the environmental approval condition and the possible production start, while B5 and B6 talk about the same area but introduce factual errors.

3.4 Atomic claim extraction

The judge evaluates factual claims rather than blindly comparing whole sentences. This makes the method more defensible when one sentence contains multiple facts.

Summary A claim units	Summary B claim units
VoltAuto announced a battery factory. The factory will be	VoltAuto announced a battery factory. The factory will be

in Mainz. The announcement happened on Monday.	in Mainz.
The project will cost €2 billion. The project will create 3,000 jobs.	The factory will cost €2 billion. The project will create 3,000 jobs.
Construction will begin in 2026.	Construction will begin in 2026.
Production could start in 2028. Production depends on environmental approval.	Environmental regulators have already approved the project. Production is guaranteed to start in 2028.

3.5 LLM judge checks each claim against the article

The LLM judge receives the article, Summary A, Summary B, and a strict factuality rubric. The judge uses the article as the source of truth and evaluates factual consistency only.

- The judge should not prefer a summary because it is longer.
- The judge should not prefer a summary because it is shorter.
- The judge should not prefer a summary because it sounds more fluent.
- The judge should ignore wording differences unless they change the factual meaning.
- The judge should allow A better, B better, tie, or unclear.

The LLM works as a structured annotator:

Article = evidence source
Summary A and Summary B = candidate outputs
Rubric = judging rules
Output = local factual judgments with reasons and confidence

The LLM judge is not treated as ground truth. It is a noisy annotator. Low-confidence, inconsistent, or unsupported judgments should be discarded or sent to audit.

3.6 Local sentence or claim-level judgment

Unit	Summary A	Summary B	Source says	Local winner
1	VoltAuto announced factory in Mainz on Monday.	VoltAuto announced factory in Mainz.	Both are supported. A includes Monday.	Tie or small A
2	Cost €2 billion and 3,000 jobs.	Cost €2 billion and 3,000 jobs.	Both are supported.	Tie
3	Construction begins in 2026.	Construction begins in 2026.	Both are supported.	Tie
4	Production could start in 2028 if approval is granted.	Approval already happened; production is guaranteed.	A is supported. B contradicts or overstates the article.	A

Summary B is not penalized because it has six sentences. It is penalized because it introduces unsupported or contradicted claims.

4. GCA Calculation for Example 1

GCA comes after the local judgments. The LLM judge produces local results, and GCA converts them into one summary-level score.

Simple scoring rule:

A wins a unit: +1
B wins a unit: -1
Tie: 0
Unclear: discard or 0

Unit	Local result	Score
Unit 1	Tie	0
Unit 2	Tie	0
Unit 3	Tie	0
Unit 4	A wins	+1

Basic GCA score:

$$\text{GCA}(A,B) = 0 + 0 + 0 + 1 = +1$$

If the margin threshold is 1, then Summary A is preferred.

A weighted GCA version can give stronger penalties to serious factual errors:

Supported tie: 0
 Minor coverage advantage: +0.5 or -0.5
 Clear contradiction: +2 or -2
 Unsupported hallucination: +2 or -2

Issue	Result	Score
Same main facts covered	Tie	0
B says approval already happened	A wins because B contradicts the article	+2
B says production is guaranteed	A wins because B overstates the article	+1

Weighted GCA score:

$$\text{GCA}(A,B) = +3$$

Final result for Example 1: Summary A is preferred. The reason is not sentence count. Summary A is preferred because it is more factually faithful to the article.

5. Example 2: Same factual information, different wording

5.1 Article

The University of Koblenz opened a new AI research lab on Friday.
 The lab will focus on trustworthy machine learning and natural language processing.
 The project is funded by a €5 million grant from the state government.

Important facts:

- F1: University of Koblenz opened a new AI research lab.
- F2: The opening happened on Friday.
- F3: The lab focuses on trustworthy machine learning.
- F4: The lab focuses on natural language processing.
- F5: The project is funded by a €5 million state government grant.

5.2 Summary A

A1: The University of Koblenz opened a new AI research lab on Friday.
 A2: The lab will study trustworthy machine learning and natural language processing.
 A3: The project is supported by a €5 million grant from the state government.

5.3 Summary B

B1: On Friday, the University of Koblenz launched a new lab for AI research.
 B2: Its work will focus on natural language processing and trustworthy machine learning.
 B3: A €5 million state government grant is funding the project.

Both summaries contain the same factual information: F1, F2, F3, F4, and F5. The wording is different, but the factual content is the same.

6. Workflow and GCA Calculation for Example 2

6.1 Sentence segmentation

Summary A: A1, A2, A3

Summary B: B1, B2, B3

6.2 Semantic alignment

A1 ↔ B1

A2 ↔ B2

A3 ↔ B3

6.3 Claim checking

Unit	Summary A	Summary B	Source support	Local winner
1	University opened AI lab on Friday.	University launched AI lab on Friday.	Both supported.	Tie
2	Trustworthy ML and NLP.	NLP and trustworthy ML.	Both supported.	Tie
3	€5 million state grant.	€5 million state grant.	Both supported.	Tie

Using the simple scoring rule:

Unit	Local result	Score
Unit 1	Tie	0
Unit 2	Tie	0
Unit 3	Tie	0

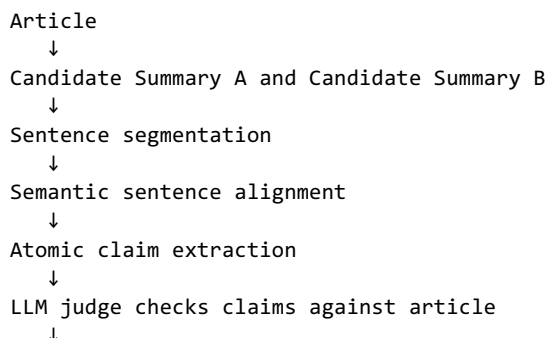
GCA score:

$$\text{GCA}(A,B) = 0 + 0 + 0 = 0$$

Final result for Example 2: no factual preference. Neither summary should be preferred for the factuality objective.

This pair should be marked as tie, no preference, or discarded from DPO training. If the judge is forced to choose between two factually equivalent summaries, it may invent weak style-based reasons. That would add label noise. For this thesis, the main target is factual reliability, so factually equivalent summaries should not create a preference pair.

7. Final Architecture Summary



```

Local factual decisions: A wins / B wins / tie / unclear
↓
GCA aggregation: sum local scores
↓
Final summary-level preference: A preferred / B preferred / no preference
↓
DPO training pair: chosen summary / rejected summary

```

8. Holistic Judging vs GCA Judging

Holistic judging	GCA judging
The judge reads the full article, Summary A, and Summary B, then gives one overall preference.	The judge evaluates aligned sentences or claims, then GCA aggregates the local decisions.
The reason for the preference can be unclear.	The reason for the preference is traceable to specific factual claims.
The judge may be influenced by length, fluency, or position bias.	The rubric forces the judge to focus on factual support from the article.
A single wrong holistic decision directly becomes noisy training data.	Unclear, low-confidence, or tied local decisions can be filtered before training.

9. Handling LLM Judge Inconsistency

The LLM judge is treated as a noisy annotator, not as ground truth. Reliability checks are needed before using its decisions for training.

```

Run 1: A first, B second → judge says A
Run 2: B first, A second → judge still says A
Result: stable label

```

If the judge changes its decision after the order is swapped, the label is unstable:

```

Run 1: judge says A
Run 2: judge says B
Result: unstable label → discard or send to human audit

```

This prevents the method from blindly trusting inconsistent LLM judgments.

10. Final Interpretation

Case	GCA result	Final action
Summary A has 4 sentences, Summary B has 6 sentences, and B contains factual errors.	Positive score for A.	Summary A is preferred and used as the chosen summary.
Summary B has better factual coverage without hallucination.	Negative score for A, or positive score for B.	Summary B is preferred and used as the chosen summary.
Both summaries contain the same factual information written differently.	Score is 0.	No preference; pair is discarded from DPO training.
Judge output is unclear or inconsistent.	Unstable or low-confidence score.	Discard or send to human audit.

The final position is that GCA does not force arbitrary choices. It only creates a training preference when there is a clear, evidence-grounded factual difference between the two summaries. If there is no factual difference, the correct output is no preference.